# Overview #2

1. **Summary of what was done in this segment:**
    a. Throughout this segment, I worked on multiple parts of the project, and by the end, I was able to achieve a model that was very accurate at hitting our goal variable. These parts involved the data loading, feature engineering, exploratory data analysis, and machine learning model development completed for predicting Integrated Primary Production (IPP) values from ocean productivity datasets. Eight datasets from various research cruises were processed and combined into a single model-ready dataframe.
2. **Overview of the dataset files used:**
    a.

| Dataset | Cruise(s) | Period | Region | Rows |
|---|---|---|---|---|
| File 1 | Albatros, Argus, NEWP, others | 1975-1994 | US East Coast + Arctic | 1,165 |
| File 2 | Eumeli | 1992 | Tropical Atlantic (W. Africa) | 71 |
| File 3 | HOTS | 1988-1989 | N. Pacific (Hawaii) | 249 |
| File 4 | EQPAC | 1992 | Equatorial Pacific | 447 |
| File 5 | BATS | 1988-1989 | N. Atlantic (Bermuda) | 254 |
| File 6 | Antiprod + others | 1977+ | Southern Ocean | 1,741 |
| File 7 | marmap + others | 1977+ | US East Coast + Atlantic | 7,329 |
| File 8 | bbs + others | 1958+ | N. Atlantic / Bermuda | 9,302 |
| **TOTAL** | | | | **20,558** |

3. **Data Cleaning:**
    a. Before continuing to work on the project, I decided to merge all the 8 files into one single dataset so that it's more abstract to work with.
    b. Once I did this, I noticed a couple of issues that had to be dealt with using the process of data cleaning. These issues include,
        i. **Invalid Month values:** 86 rows contained out-of-range Month values (e.g., Month = 84) and were dropped.
        ii. **Invalid date combinations:** 26 rows contained impossible dates (e.g., February 30) and were removed using errors='coerce'
        iii. **Missing Longitude:** Files 3, 4, 5, and 7 had missing LONG values, limiting geographic visualization but not affecting model training.
        iv. **IPP recorded once per station:** IPP is only present at the surface row of each station, which is expected by design.
    **c. Final clean master dataframe: 20,446 rows**
4. Feature Engineering:
    a. As per the recommended guidelines, six features were engineered from the raw dataset.

b. These Features include:
    i. **Day of year:** The Day and Month columns were combined into a single DayOfYear feature representing the sequential day of the year (1-365). A fixed dummy year of 2001 was used to ensure consistency across all datasets regardless of the actual recording year.
    ii. **Temp, DayLength, Irr:** These three columns were used directly as individual features after numeric conversion and NaN removal.
    iii. **Zopt(Euphotic Zone Depth):** Zopt represents the depth at which solar irradiance drops to 1% of the surface value. The bottom boundary of the productive zone where phytoplankton can perform photosynthesis. It was derived by joining the Depth and %Irr columns for each station and interpolating linearly between the depth rows immediately above and below the 1% threshold.

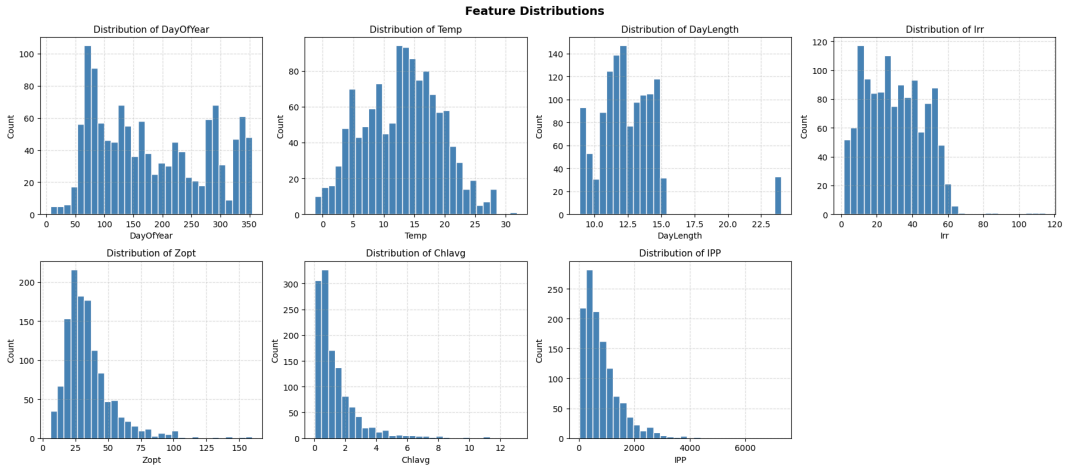
| Statistic | Value |
|---|---|
| Stations with Zopt computed | 2,358 / 2,486 (94.8%) |
| Stations with NaN (Arctic/Antarctic winter) | 128 |
| Mean Zopt | 51.3 m |
| Min Zopt | 3.0 m |
| Max Zopt | 282.0 m |

    iv. **Chlavg(Average Chlorophyll within Euphotic Zone):** Chlavg represents the average chlorophyll concentration from the surface down to Zopt. Chlorophyll is a proxy for phytoplankton biomass; higher chlorophyll means more phytoplankton present.

5. **Final dataset on which the model will train:**
    a. After merging all engineered features and dropping rows with any missing values, the final dataset contained **1,244 rows**, each representing one unique sampling station.

| Feature | Description | Type |
|---|---|---|
| DayOfYear | Sequential day of year (1-365) | Engineered |
| Temp | Sea surface temperature (degC) | Direct |
| DayLength | Solar hours per day | Direct |
| Irr | Surface solar irradiance | Direct |
| Zopt | Euphotic zone depth (m) | Engineered |
| Chlavg | Mean chlorophyll within euphotic zone | Engineered |
| **IPP (TARGET)** | **Integrated Daily Carbon Fixation (mgC/m2/day)** | **Target** |

6. EDA:
    a. Feature Distribution:
        i. Histograms were plotted for all six features and the IPP target variable




ii. Observations made from graphs:

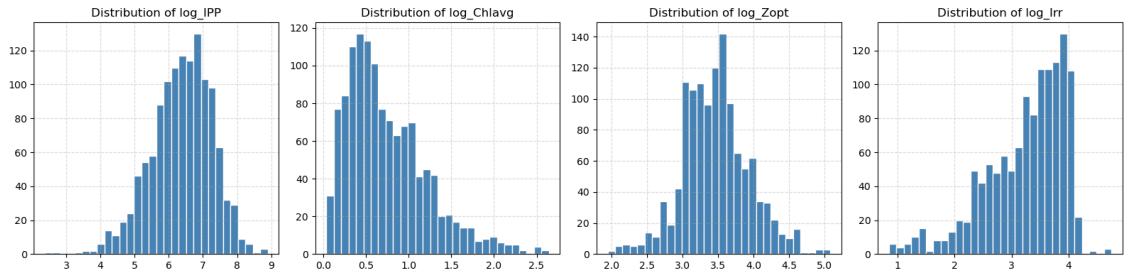
1. Zopt, Chlavg, Irr, and IPP were all right-skewed
2. DayOfYear was reasonably spread across the full year with a slight peak in early spring (days 50-100).
3. Temp peaked in the 15-20 degC range, consistent with mid-latitude sampling bias.

b. Log Transformation:

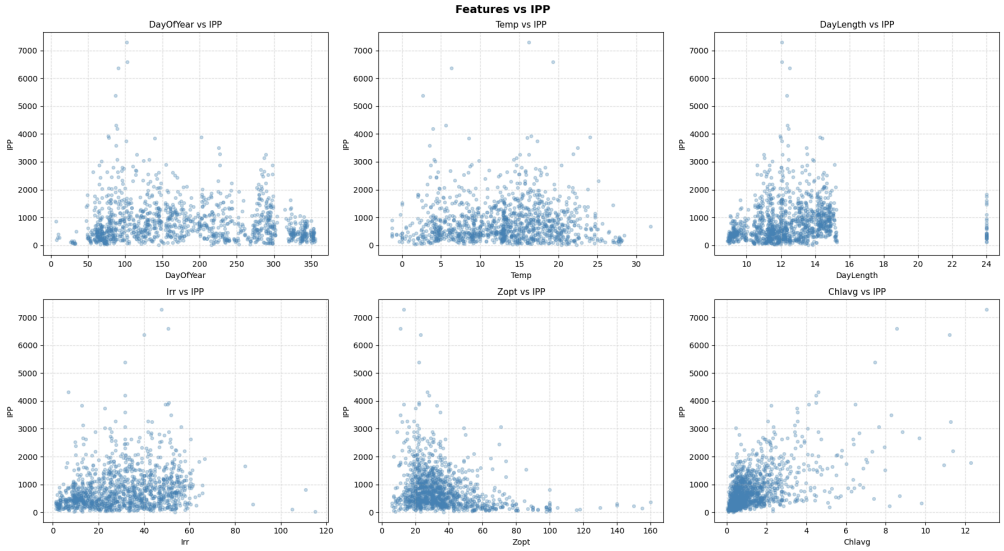
i. Due to the right-skewed nature of IPP, Chlavg, Zopt, and Irr, log transformations were applied using np.log1p().

ii. I did this because of three main reasons:

1. To reduce the influence of extreme outliers on model training.
2. To normalize the distributions and bring all values into a comparable range.
3. Because IPP is naturally a multiplicative relationship, log(IPP) = log(Chlavg) + log(Zopt) + log(Irr) + ..., making log-space more appropriate for tree-based models.



c. Feature vs IPP Scatter Plots:

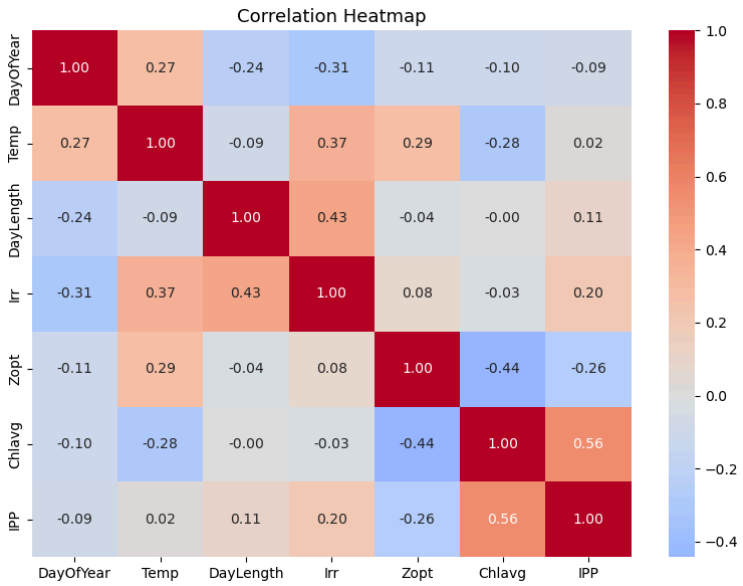



i. Chlavg showed the clearest positive relationship with IPP.

ii. Low Zopt always corresponded to low IPP.

iii. Temp and DayOfYear showed no clear linear trend with IPP individually.

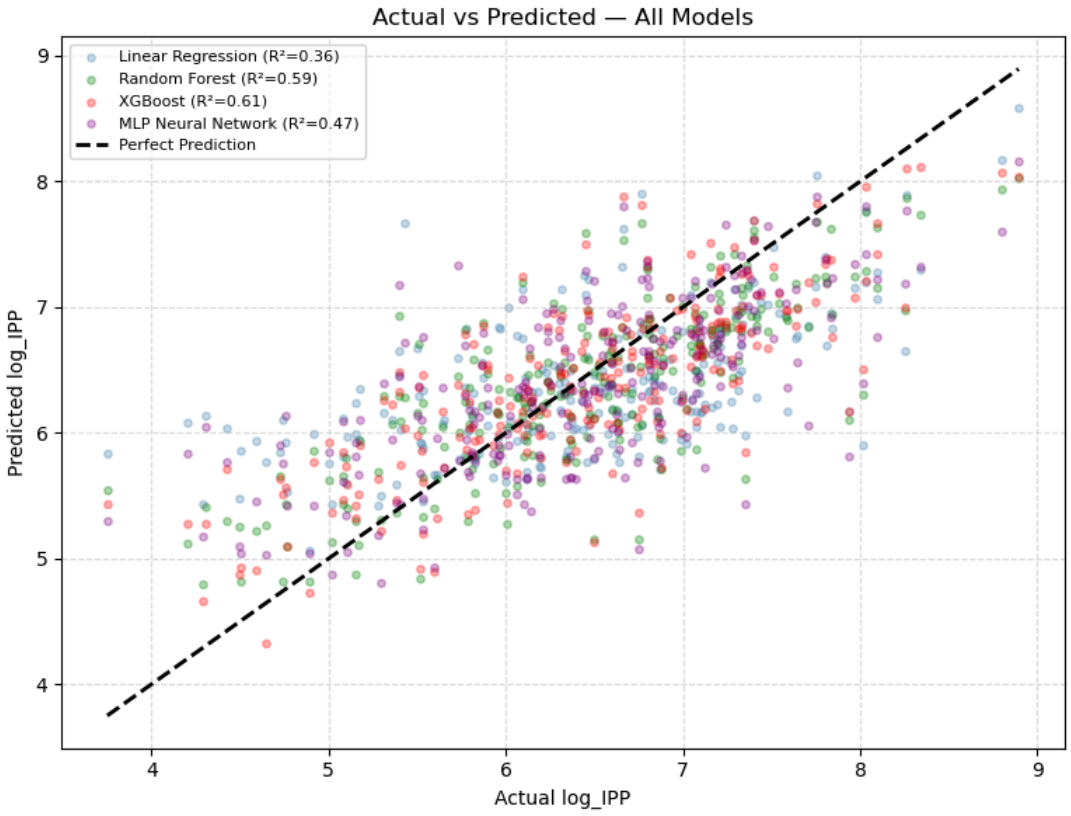
d. Correlation Heatmap:



i.

| Feature | Correlation with IPP | Interpretation |
|---|---|---|
| log_Chlavg | 0.56 | Strongest predictor |
| log_Irr | 0.20 | Weak positive |
| DayLength | 0.11 | Weak positive |
| log_Zopt | -0.26 | Shallow zone = less productive |
| Temp | 0.02 | Almost no direct correlation |
| DayOfYear | -0.09 | Almost no direct correlation |

7. Model Training:
    a. Preprocessing:
        i. Standard Scaling was applied after the train/test split. The scaler was fit only on training data and applied to test data to prevent data leakage.
        ii. Train/Test Split: 80% training (995 rows), 20% testing (249 rows), random_state=42.
    b. Models Trained:



```
=== Linear Regression ===
R² Score : 0.3593
RMSE     : 0.7184

=== Random Forest ===
R² Score : 0.5927
RMSE     : 0.5727

=== XGBoost ===
R² Score : 0.6092
RMSE     : 0.5611

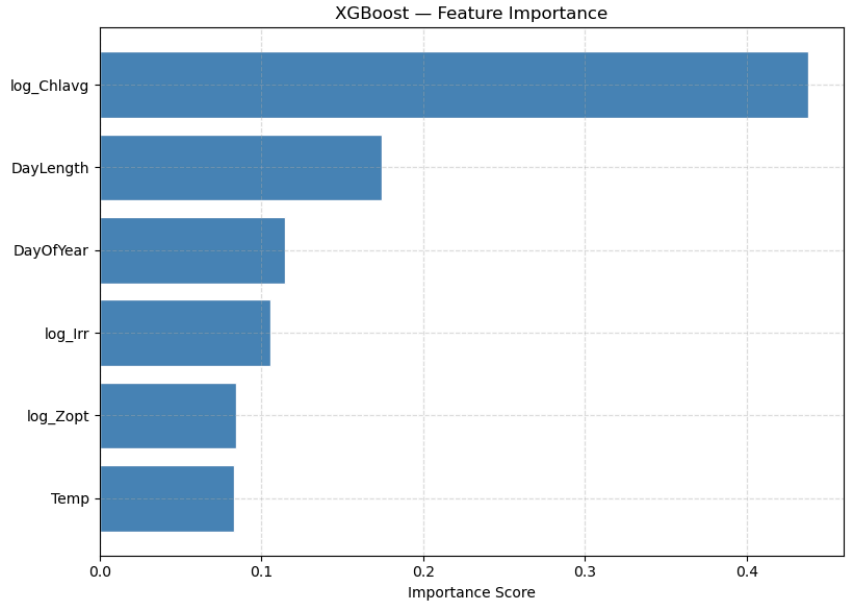
=== MLP Neural Network ===
R² Score : 0.4708
RMSE     : 0.6528

============= MODEL COMPARISON =============
Model                          R²     RMSE
-------------------------------------------
Linear Regression          0.3593   0.7184
Random Forest              0.5927   0.5727
XGBoost                    0.6092   0.5611
MLP Neural Network         0.4708   0.6528

Best Model: XGBoost (R² = 0.6092)
```

i. XGBoost was selected as the best-performing model. Linear Regression performed weakest, confirming the non-linear nature of IPP relationships. The MLP Neural Network underperformed relative to XGBoost, likely due to the small dataset size (1,244 rows) being insufficient for deep learning to reach its full potential.

8. XGBoost Feature importance:
    a. log_Chlavg was by far the most important feature, consistent with its correlation of 0.56 with IPP from the heatmap.




9. Hyperparameter Tuning:

a. RandomizedSearchCV was used with 5-fold cross-validation to optimize XGBoost hyperparameters. Since I am working on a CPU environment, RandomizedSearchCV was preferred over GridSearchCV for efficiency. Trying 150 random combinations rather than exhaustively testing every possible combination.
b. Best parameters found:

| Parameter | Default Value | Tuned Value | Effect |
|---|---|---|---|
| n_estimators | 100 | 300 | More trees = better learning |
| max_depth | 4 | 6 | Slightly deeper trees |
| learning_rate | 0.1 | 0.05 | Smaller steps = more careful |
| subsample | 1.0 | 0.7 | 70% data per tree = less overfit |
| colsample_bytree | 1.0 | 0.7 | 70% features per tree |
| min_child_weight | 1 | 3 | Prevents splitting on noise |
| gamma | 0 | 0.1 | Adds regularization |

c. Tuning Results:
    i. I did tuning in three different ways, and the table below shows the difference from a base XGBoost model to a more modified/tuned model.

| Model | R2 Score | RMSE | Improvement |
|---|---|---|---|
| XGBoost Default | 0.6092 | 0.5611 | Baseline |
| XGBoost Tuned (80 iter) | 0.6252 | 0.5494 | +0.0160 |
| **XGBoost Tuned (150 iter)** | **0.6455** | **0.5344** | **+0.0363** |

10. Concluding thoughts:
    a. Final Model performance tabularly:

| Metric | Value |
|---|---|
| Model | XGBoost (Tuned) |
| R2 Score | 0.6455 |
| RMSE (log scale) | 0.5344 |
| Most Important Feature | log_Chlavg (44% importance) |

    b. Key Findings:
        i. Chlorophyll average within the euphotic zone (Chlavg) is the dominant predictor of IPP.
        ii. Temperature and Day of Year showed minimal direct correlation with IPP, though XGBoost captured indirect non-linear effects.

iii. The dataset size of 1,244 stations after feature engineering limited model performance; more data would likely improve R2 significantly.
iv. Log transformation of skewed features (IPP, Chlavg, Zopt, Irr) significantly improved model training by reducing outlier influence.

c. Possible next steps:
i. Adding more datasets to increase training samples beyond 1,244.
ii. Exploring additional features.
iii. Testing ensemble methods combining XGBoost and Random Forest predictions.